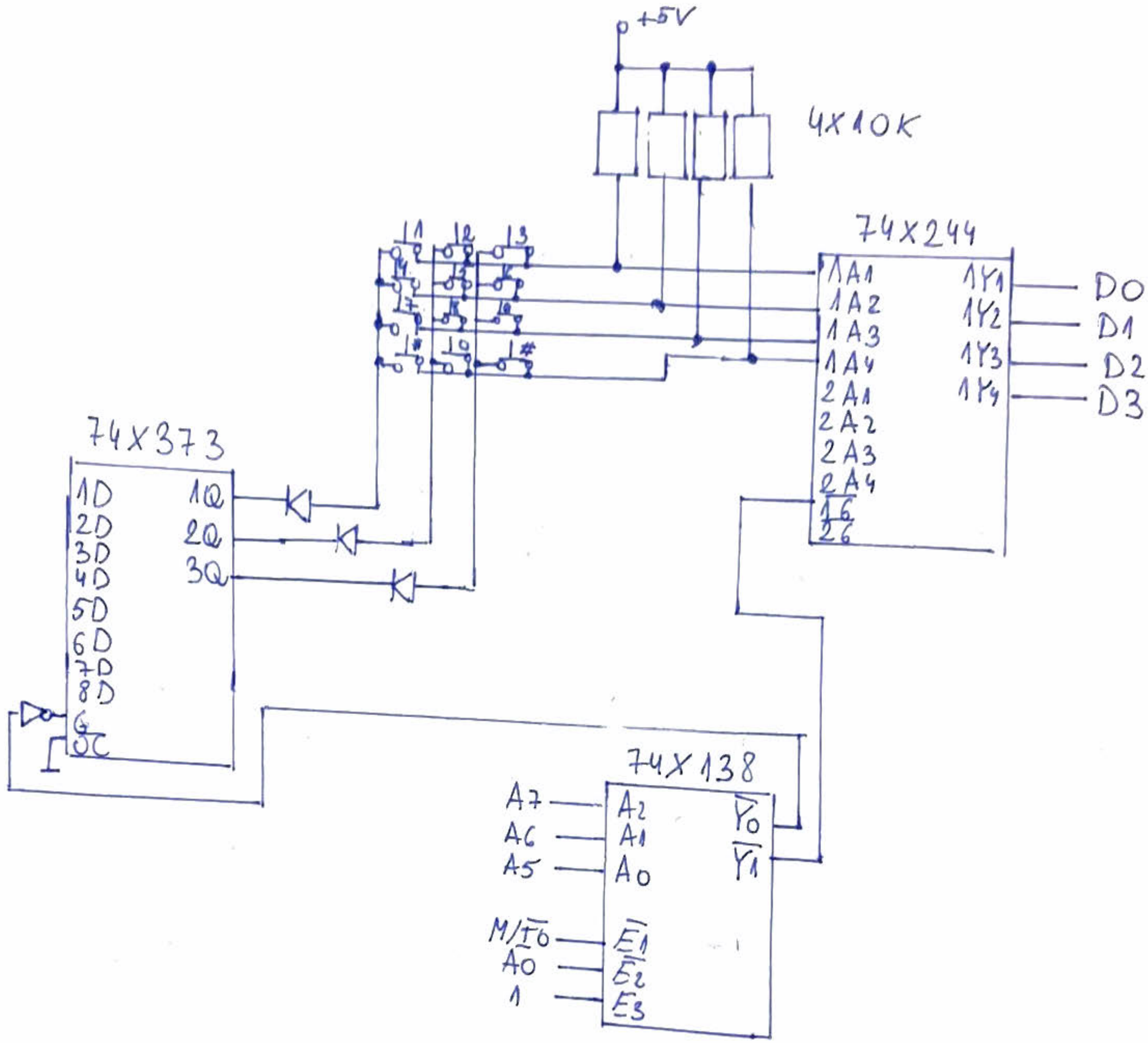
8. Conectarea la o unitate centrală o tastatură cu 12 caractere.

Rezolvare:

Schema:


+5V
4X10K
74X244
1A1
1A2
1A3
1A4
2A1
2A2
2A3
2A4
1G
2G
1Y1
1Y2
1Y3
1Y4
D0
D1
D2
D3
74X373
1D
2D
3D
4D
5D
6D
7D
8D
G
OC
1Q
2Q
3Q
74X138
A7
A6
A5
M/IO
A0
1
A2
A1
A0
E1
E2
E3
Y0
Y1

Programul:

; punem 0 pe prima coloană și verificăm dacă s-au atins tastele: 1, 4, 7, *.

7	6	5	4	3	2	1	0	
1	1	1	1	1	1	1	0	=> FEH

```
REIA: MOV AL,0FEH
OUT 00H, AL
IN AL, 20H
AND AL, 01H;    | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
JZ TASTA1
IN AL, 20H
AND AL, 02H;    | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
JZ TASTA4
IN AL, 20H
AND AL, 04H;    | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
JZ TASTA7
IN AL, 20H
AND AL, 08H;    | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
JZ TASTA*
```

; punem 0 pe coloana a doilea și verificăm dacă s-au atins tastele: 2, 5, 8, 0

1	1	1	1	1	1	0	1	=> FDH

```
MOV AL, 0FDH
OUT 00H, AL
IN AL, 20H
AND AL, 01H;   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
JZ TASTA2
IN AL, 20H
AND AL, 02H;   | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
JZ TASTA5
IN AL, 20H
AND AL, 04H;   | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
JZ TASTA8
IN AL, 20H
AND AL, 08H;   | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
JZ TASTA0
```

i punem 0 pe coloana a treia și verificăm dacă s-au atins tastele: 3, 6, 9, #

1	1	1	1	1	0	1	1

=> FBH

```
MOV AL, 0FBH
OUT 00H, AL
IN AL, 20H
AND AL, 01H;   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
JZ TASTA3
IN AL, 20H
AND AL, 02H;   | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
JZ TASTA6
IN AL, 20H
AND AL, 04H;   | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
JZ TASTA9
IN AL, 20H
AND AL, 08H;   | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
JZ TASTA#
```